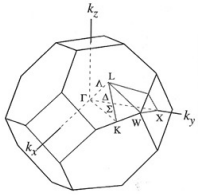
Introduction to Quantum Mechanics

Baile Chen

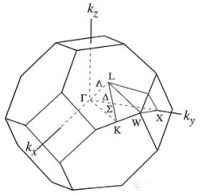
School of Information Science and Technology

Shanghai Tech University



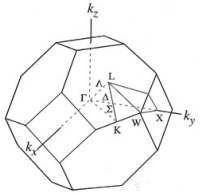
History

- Around 1900: defined as “begin of modern physics”
- What was known in physics:
  - Newton’s law, mechanics
  - Thermodynamics: heat flow, efficiency
  - Chemistry: ideal gas flow, stat mechanics
  - Maxwell equations: waves
  - Optics: lenses, prisms, gratings, spectroscopy
  - Sir Joseph John Thomson: electron
  - Wilhelm Röntgen: X-rays



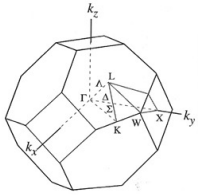
A Couple of Issues Left

- Ultraviolet Catastrophe for Blackbody Radiation
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  - ➢Experiment:
    - ✓Above critical wavelength: no electrons
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- Atom will not be stable

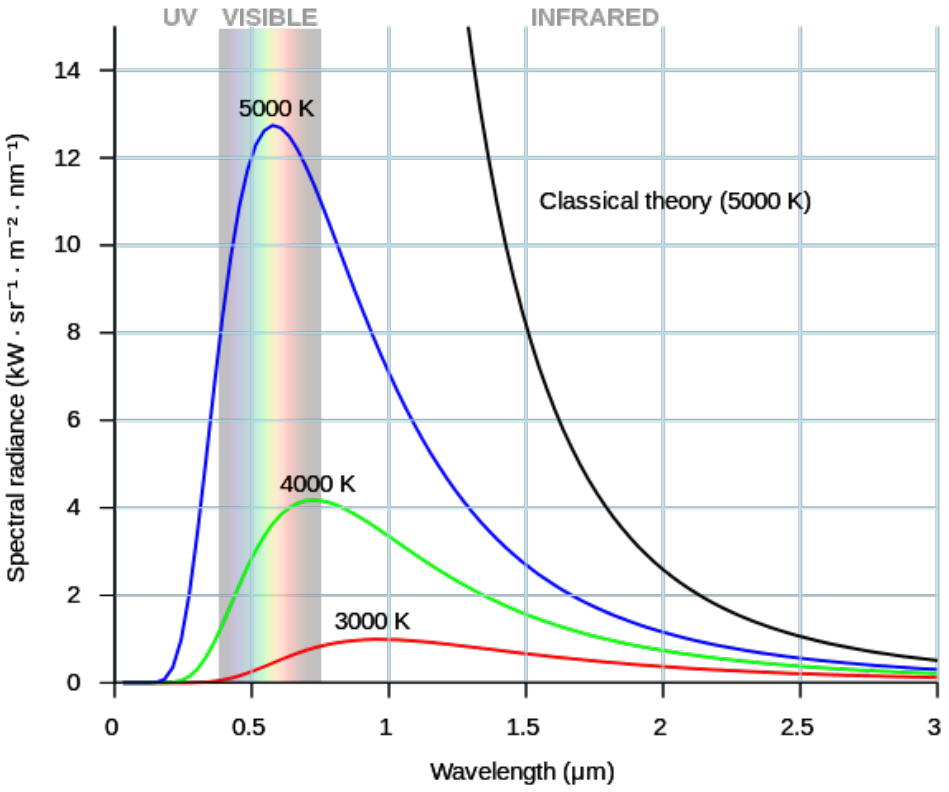


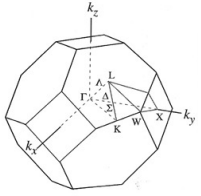
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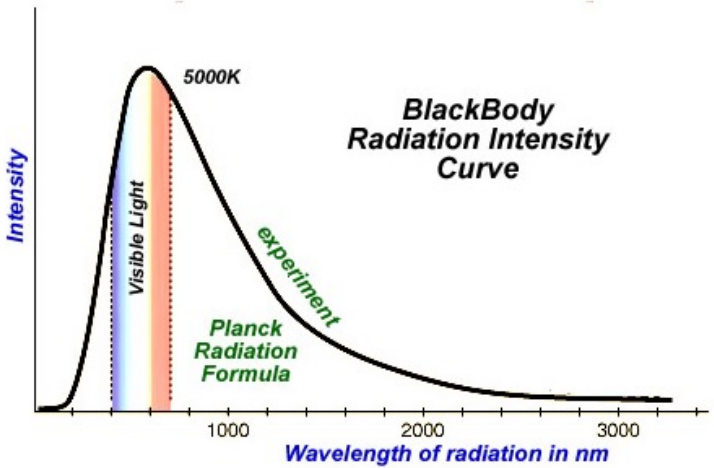


Blackbody Radiation





Ultraviolet Catastrophe



Max Planck devised a solution to the **Ultraviolet Catastrope** problem that required the quantization of energy. Energy comes in packets called quanta.

$$I(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda kT}} - 1}$$

$E = hf = \frac{h\nu}{2\pi} = \hbar\nu$

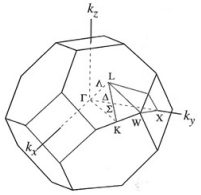
$f = \frac{\nu}{2\pi}$

1901

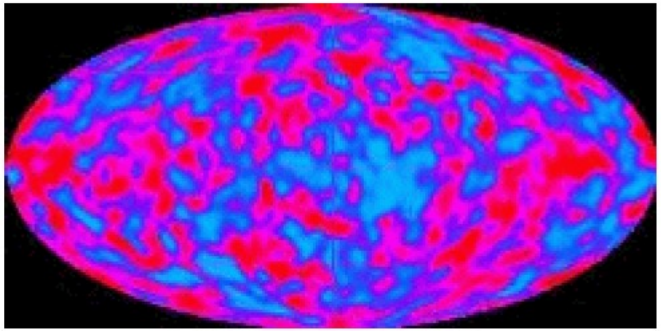
EM emission occurs in discrete quanta of

E = nhf n=1,2,3,….

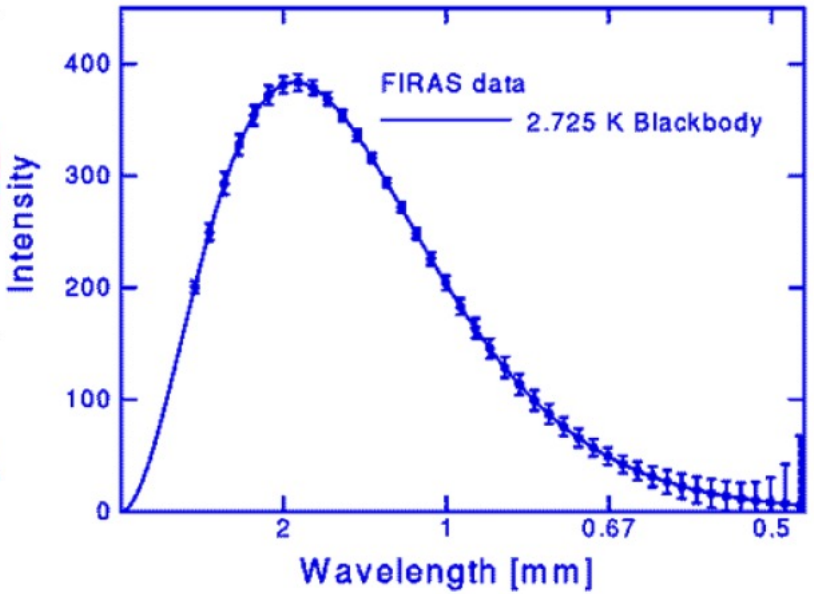
h is Plank's constant, f is the oscillator frequency

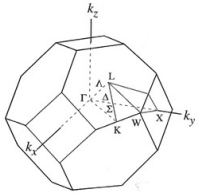


Recent Example: COBE Data



J.C. Mather, Astrophysics J., 1990.

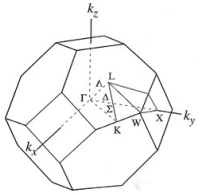




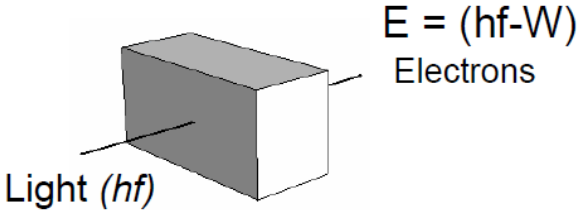
Show that the cosmic background temperature is approximately 3K. Can you “see” this radiation?

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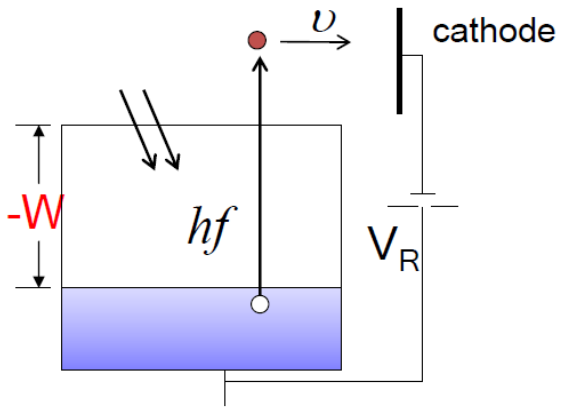
Photoelectric Effect

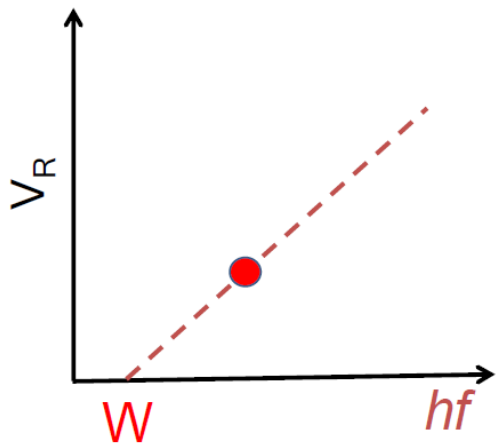


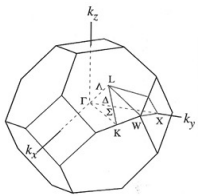
1905

- Einstein explained the photoelectric effect
- Photo energy: E = hυ
- Light sometimes acts as particle

$$qV_R \approx \left(\tfrac{1}{2}\right) m_0 \upsilon^2 = hf - W$$

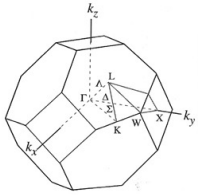






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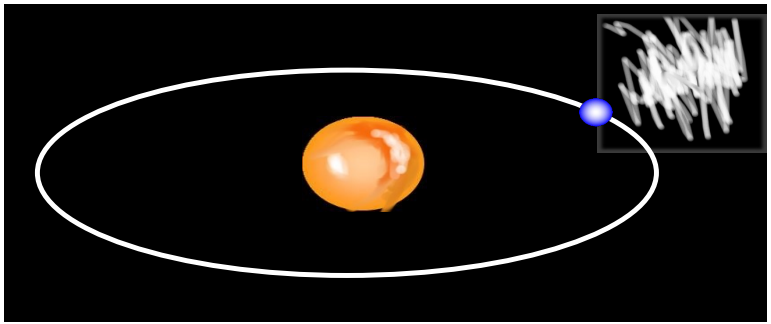


Solar System Model of Atom

$$mv^2/r = Zq^2/4\pi\varepsilon_0 r^2$$

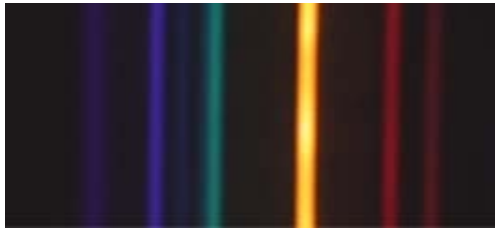
Centripetal force

Electrostatic force

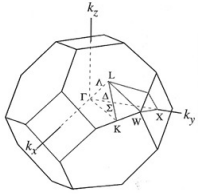


Continuous radiation from orbiting electron

- 1910
- Pb1: Atom would be unstable! (expect nanoseconds observe billion years!)
- Pb2: Spectra of atoms are discrete!



Spectrum of Helium



Transitions $E_0(1/n^2 - 1/m^2)$ (n,m: integers)

Why Waves?

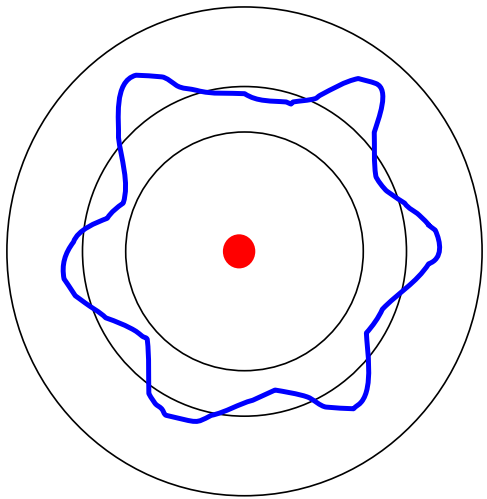
- Only certain modes allowed (like a plucked string)
- $n\lambda = 2\pi r$ (fit waves on circle)
- Momentum ~ 1/wavelength **(DeBroglie, 1924)**
- $p = mv = h/\lambda$ (massive classical particles → vanishing λ)
- This means angular momentum is quantized $mvr = nh/2\pi = n\hbar$

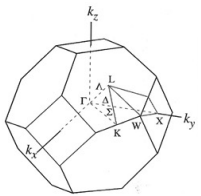
$$\frac{mv^2}{r} = \frac{zq^2}{4\pi\varepsilon_0 r^2} \Rightarrow mv^2 r = \frac{zq^2}{4\pi\varepsilon_0}$$

$$n\lambda = 2\pi r \Rightarrow \frac{nh}{p} = 2\pi r$$

$$\Rightarrow \frac{nh}{mv} = 2\pi r$$

$$\Rightarrow v_n = \frac{zq^2}{4\pi\varepsilon_0 n\hbar}$$

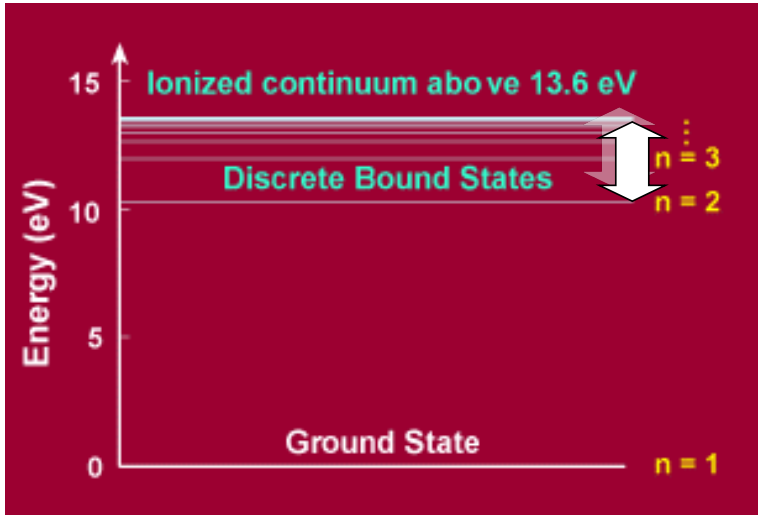




From 2 equations, $r_n = (n^2/Z)\, a_0$

$a_0 = h^2\varepsilon_0/\pi q^2 m = 0.529$ Å (Bohr radius)

Bohr's Suggestion



$E = mv^2/2 - Zq^2/4\pi\varepsilon_0 r$

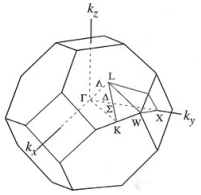
Using previous two equations

$E_n = (Z^2/n^2)E_0$

$E_n = -\frac{1}{2} m \left(\frac{Zq^2}{4\pi\varepsilon_0\hbar}\right)^2 \frac{1}{n^2}$

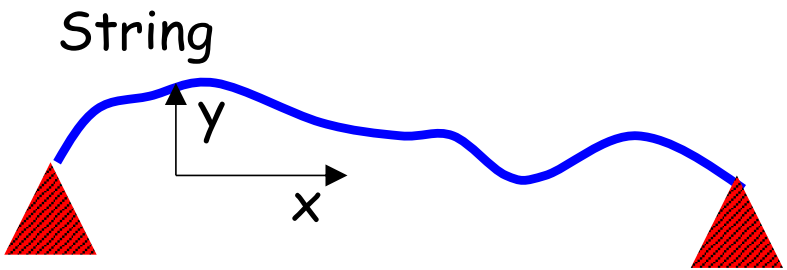
$E_0 = -mq^4/8h^2\varepsilon_0^2 = -13.6$ eV
$= 1$ Rydberg

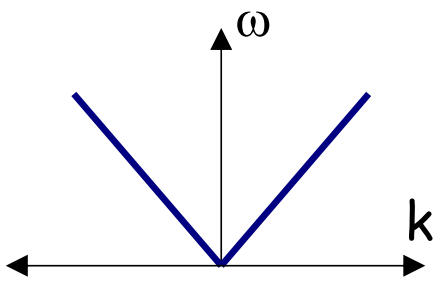
- 1913
- Transitions $E_0(1/n^2 - 1/m^2)$ (n,m: integers)
- Explains discrete atomic spectra
- So need a suitable Wave equation so that imposing boundary conditions will yield the correct quantized solutions



What Should Our Wave Equation Look Like?

$\partial^2 y/\partial t^2 = v^2(\partial^2 y/\partial x^2)$



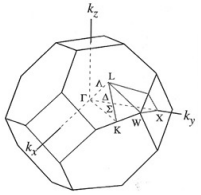


Solution: $y(x,t) = y_0 e^{i(kx-\omega t)}$

$\omega^2 = v^2 k^2$

k 波数 $= \frac{2\pi}{\lambda}$

What is the dispersion (ω-k) for a particle?

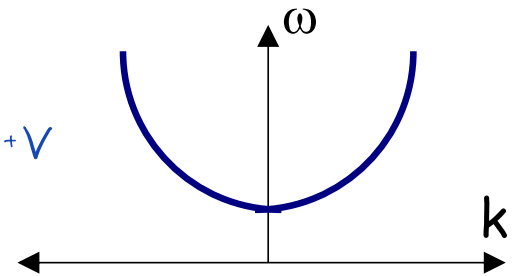


What Should Our Wave Equation Look Like?

Quantum theory:

- $E=hf = \hbar\omega$ (Planck's Law)
- $p = h/\lambda = \hbar k$ (de Broglie Law) $p = \frac{h}{\lambda} = \frac{h}{\frac{2\pi}{k}} = \hbar k$
- $E = p^2/2m + U$ (energy of a particle)

$E = E_k + E_p \Rightarrow \hbar\omega = \frac{p^2}{2m} + V \Rightarrow \hbar\omega = \frac{\hbar^2 k^2}{2m} + V$

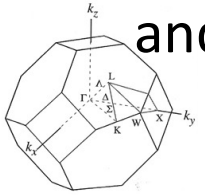


Thus, dispersion we are looking for is

$$\omega \propto k^2 + U$$

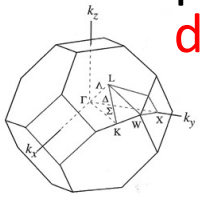
So we need <u>one</u> time-derivative and <u>two</u> spatial derivatives

$$\partial^2 y/\partial t^2 = v^2(\partial^2 y/\partial x^2)$$



Schrödinger Equation

- Formulated wave mechanics
- Wavefunction Ψ to describe the system
  - $-\frac{\hbar^2}{2m}\nabla^2\Psi(\boldsymbol{r},t)+U(\boldsymbol{r},t)\Psi(\boldsymbol{r},t)=i\hbar\frac{\partial\Psi(\boldsymbol{r},t)}{\partial t}$
  - U: potential energy, i.e. electrostatic potential
  - ∇^2:Laplace operator
  - $\hbar\equiv\frac{h}{2\pi}$
- If U is time-independent: $U(\boldsymbol{r},t)=U(\boldsymbol{r})$ 势. 假设不与空间相关.
  - $\Psi(\boldsymbol{r},t)=\varphi(\boldsymbol{r})\emptyset(t)=\varphi(\boldsymbol{r})e^{-i(\frac{E}{\hbar})t}$
  - $-\frac{\hbar^2}{2m}\nabla^2\varphi_n(\boldsymbol{r})+U(\boldsymbol{r})\varphi_n(\boldsymbol{r})=E_n\varphi_n(\boldsymbol{r})$
- Physical meaning of wave function: $|\Psi(\boldsymbol{r},t)|^2=|\varphi(\boldsymbol{r})|^2$ is probability density function



$U(x,t) = U(x)$.

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + U(x)\psi$$

$$\psi(x,t) = \varphi(x)\cdot a(t)$$

$$i\hbar\frac{a'(t)}{a(t)} = -\frac{\hbar}{2m}\cdot\frac{\varphi''(x)}{\varphi(x)} + U(x) = E$$

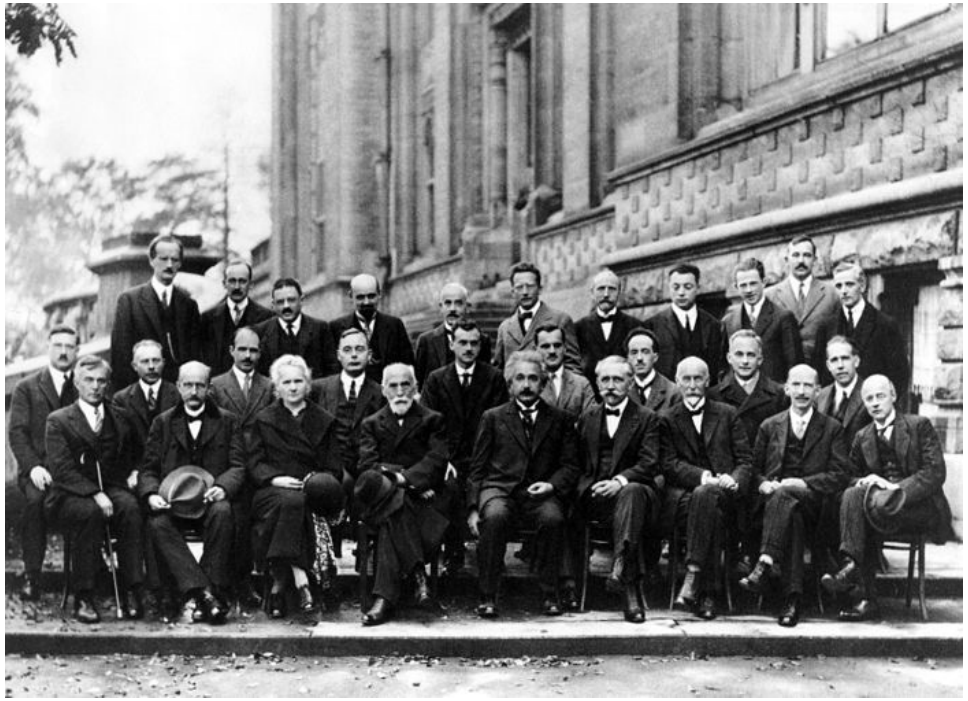
$i\hbar\frac{a'(t)}{a(t)}$：与 x 无关；$-\frac{\hbar}{2m}\cdot\frac{\varphi''(x)}{\varphi(x)} + U(x)$：与 t 无关

E：待定常数

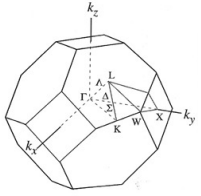
① $i\hbar\frac{a'(t)}{a(t)} = E \Rightarrow a(t) = e^{-i\frac{E}{\hbar}t} \Rightarrow a_n(t) = e^{-i\frac{E_n}{\hbar}t}$

② $-\frac{\hbar}{2m}\frac{\varphi''}{\varphi} + U(x) = E$

Fifth Conference Participants, 1927.



A. Piccard, E. Henriot, P. Ehrenfest, E. Herzen, Th. de Donder, E. Schrödinger, J.E. Verschaffelt, W. Pauli, W. Heisenberg, R.H. Fowler, L. Brillouin;
P. Debye, M. Knudsen, W.L. Bragg, H.A. Kramers, P.A.M. Dirac, A.H. Compton, L. de Broglie, M. Born, N. Bohr;
I. Langmuir, M. Planck, M. Skłodowska-Curie, H.A. Lorentz, A. Einstein, P. Langevin, Ch.-E. Guye, C.T.R. Wilson, O.W. Richardson

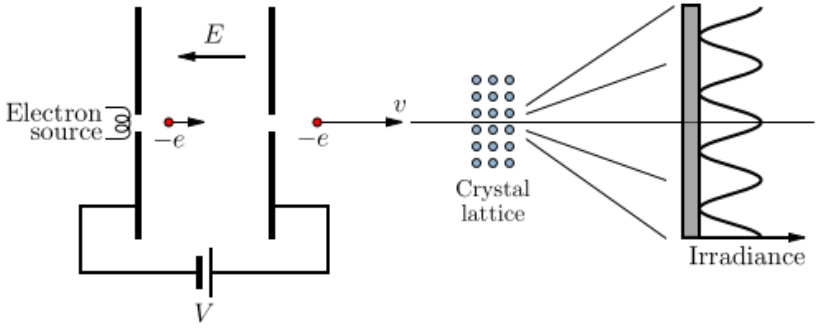


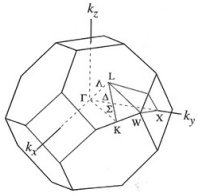
Wave-particle Duality

- Soccer ball
  - m=0.45kg
  - v=40m/s
  - p=mv=18 kg m/s
  - $\lambda = \frac{h}{p} = ???$
  - Size of goal???



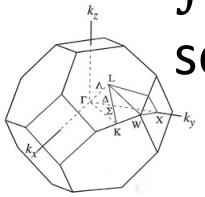
- Electron moving across potential difference
  - $m = 9.1\times10^{-31} kg$
  - E=1000eV
  - $p = \sqrt{2mE}$
  - $\lambda = \frac{h}{p} = ???$





Schrödinger Rules

- Wave-particle duality
  - ➢λ is the same scale as the system size: wave
  - ➢λ ≪system size: particle
- Particles are described by wavefunction Ψ
- Ψ is a complex solution to SWE
- Ψ are finite, continuous
- $\Psi\Psi^* dV$: probability of finding the particle inside dV.
  - ➢$\Psi\Psi^*$ is real
- $\int \Psi\Psi^* dV = 1$ normalization, probability to find the particle somewhere in V is 1



More Schrödinger Rules

- Expectation values: most likely values for position, momentum, energy,…
- $\langle x \rangle = \int \Psi^* x \Psi dV$:expected position of particle $\quad \hat{x} \rightarrow x$
- $\langle p \rangle = \int \Psi^* \frac{\hbar}{i} \frac{\partial \Psi}{\partial x} dV$: momentum $\quad \hat{p} \rightarrow \frac{\hbar}{i} \frac{\partial}{\partial x}$
- Commutation relation: $[\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar$
- Operators are like matrices

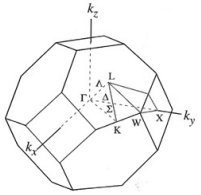
位置的期望：$\langle x \rangle = \int |\psi|^2 x \, dx = \int \psi^* x \psi \, dx$

$\psi(x) \xleftrightarrow{\text{Fourier Transform}} \psi(p)$

Operator : $\hat{x} \rightarrow x$. $\quad \hat{p} \rightarrow -i\hbar \frac{\partial}{\partial x}$

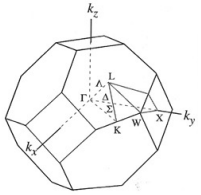
$[\hat{x}, \hat{p}]\psi(x) = x \cdot (-i\hbar \frac{\partial}{\partial x}) \cdot \psi(x) + i\hbar \frac{\partial}{\partial x}(x \cdot \psi(x))$

$= i\hbar \cdot \psi(x)$



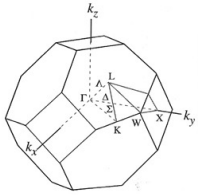
The Uncertainty Principle

- 1927 (Heisenberg)
- $\Delta p \Delta x \geq \hbar/2$
- $\Delta E \Delta t \geq \hbar/2$
- That means:
  - You can't simultaneous measure position and momentum accurately
  - You can't simultaneous measure energy and time accurately
- $\hbar$ is small, so uncertainty principle is only significant for subatomic particles



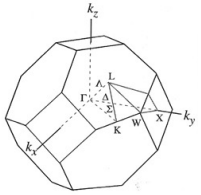
Time-independent Case

- 1D case: If U(x, t) = U(x)
  - $\Psi(x,t) = \varphi(x)\emptyset(t) = \varphi(x)e^{-i(\frac{E}{\hbar})t}$
  - $-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\varphi_n(x) + U(x)\varphi_n(x) = E_n\varphi_n(x)$
  - $\Psi(x,t) = \sum_n C_n\varphi_n(x)\, e^{-i(\frac{E_n}{\hbar})t}$
  - C_n is determined by the initial condition of wavefunction at t=0.



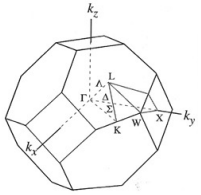
Use Schrödinger Equation (SE) to Describe a Free Particle

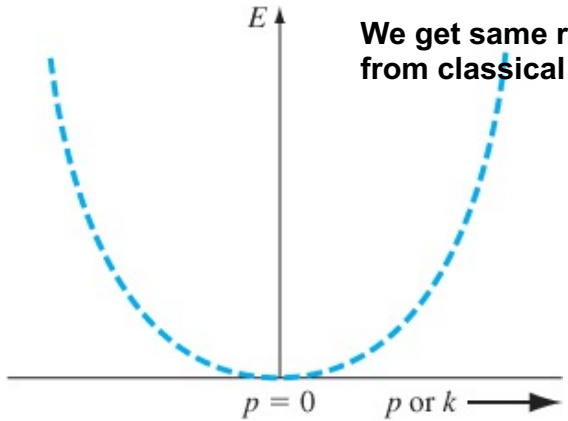
- Specially case: U(x)=0
  - $-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\varphi(x) = E\varphi(x)$
  - $\varphi(x) \sim e^{\pm ikx}$
  - $\langle p \rangle = \int \varphi^* \frac{\hbar}{i}\frac{\partial \varphi}{\partial x} dx = \int \varphi^* \hbar k \varphi dx = \hbar k$
  - $E = \frac{\hbar^2 k^2}{2m}$
  - If you know E(k), how to solve for m?
    - ✓ Electron in crystal: treat them as "free particle", but with a different mass



Example 1: Free Space

- $\frac{d^2}{dx^2}\psi(x) + \frac{2m}{\hbar^2}(E - V(x))\psi(x) = 0$
- Solutions for spatial part if V(x) = const. = 0:
  - $\psi(x) = Ae^{ikx} + Be^{-ikx}$ with wavenumber: $k = \sqrt{\frac{2mE}{\hbar^2}} = \frac{p}{\hbar} = \frac{2\pi}{\lambda}$
- Forward and backward traveling waves
- Total solution:
  - $\Psi(x,t) = Ae^{i(kx-\frac{E}{\hbar}t)} + Be^{-i(kx+\frac{E}{\hbar}t)}$
  - Plot E vs. k: $E = \frac{k^2\hbar^2}{2m}$



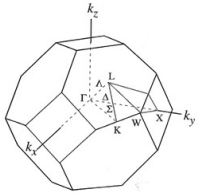


We get same result from classical physics:

Free particle: all values allowed.

Example 1: Free Space

- Probability to find particle in dx assuming it travels in one direction (B=0):
  - $\Psi(x,t)^* \Psi(x,t)dx = |\Psi(x,t)|^2 dx = A^* A\, dx$
- Constant probability for all x.
- Problem: Normalization. Integration of probability density over all space is not 1 – since wavefunction extends to infinity.
  - Use wave packet (superposition of waves with different wavelengths) to localize particle.
  - or, limit space to some large, but finite value.



Example 2

- Particle in an infinite potential well or particle in a box

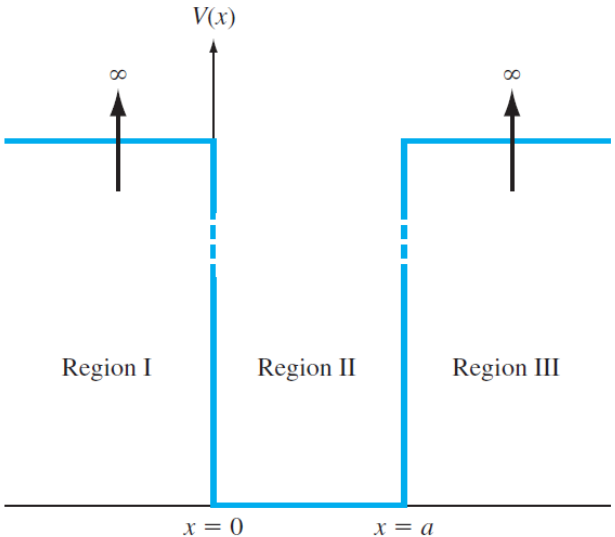


Figure 2.6 | Potential function of the infinite potential well.

$$\frac{d^2\psi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - V(x))\psi(x) = 0$$

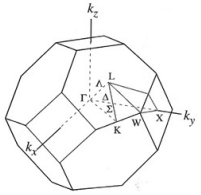
$x<0,\ x>a,\ V(x)=\infty \Rightarrow E=0$

$$0<x<a,\quad \frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0$$

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

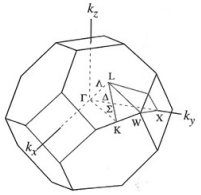
$$k = \frac{\sqrt{2mE}}{\hbar}$$

$$\psi(0) = \psi(a) = 0 \qquad \begin{cases} B = 0 \\ A\sin(ka) = 0 \end{cases}$$



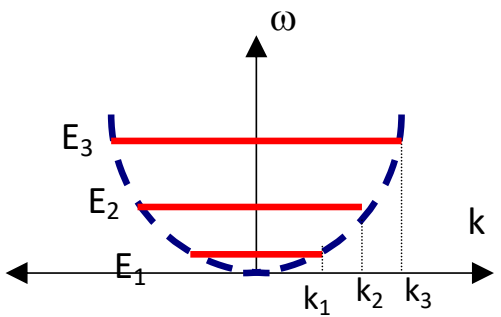
Example 2

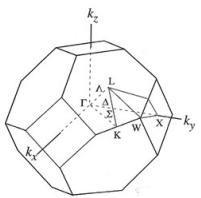
- Solution:
- Region I and III: ψ=0
- Region II:
  - $\psi(x) = A'e^{ikx} + B'e^{-ikx} = Asinkx + Bcoskx$
- Boundary condition
  - ψ(0)= 0
    - ✓ $\psi(x) = Asinkx$
  - ψ(a)=0
    - ✓ $\psi(a) = Asinka$=0
    - ✓ $k_n = \frac{n\pi}{a}$



Example 2

- $E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2 n^2 \pi^2}{2ma^2}$ n=1,2,3…
- $E_n = n^2 E_1$
- the E vs k plot looks like?
- Wavefunction looks like?
- Probability to find the particle?





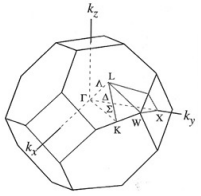
Example 2

- $E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2 n^2 \pi^2}{2ma^2}$ n=1,2,3…
- $E_n = n^2 E_1$
- the E vs k plot looks like?
- Wavefunction looks like?
- Probability to find the particle?

$k_n = \frac{n\pi}{a}$

$\lambda_n = \frac{2\pi}{k_n}$

$\Rightarrow n \cdot \frac{\lambda_n}{2} = a$



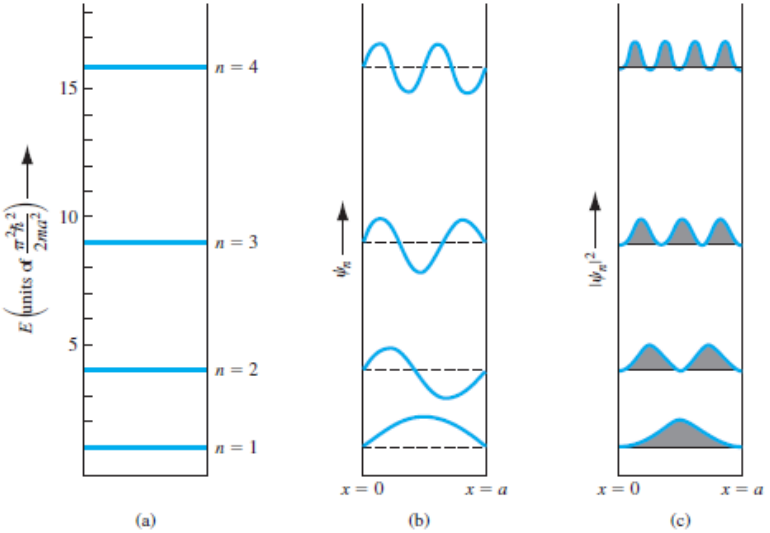
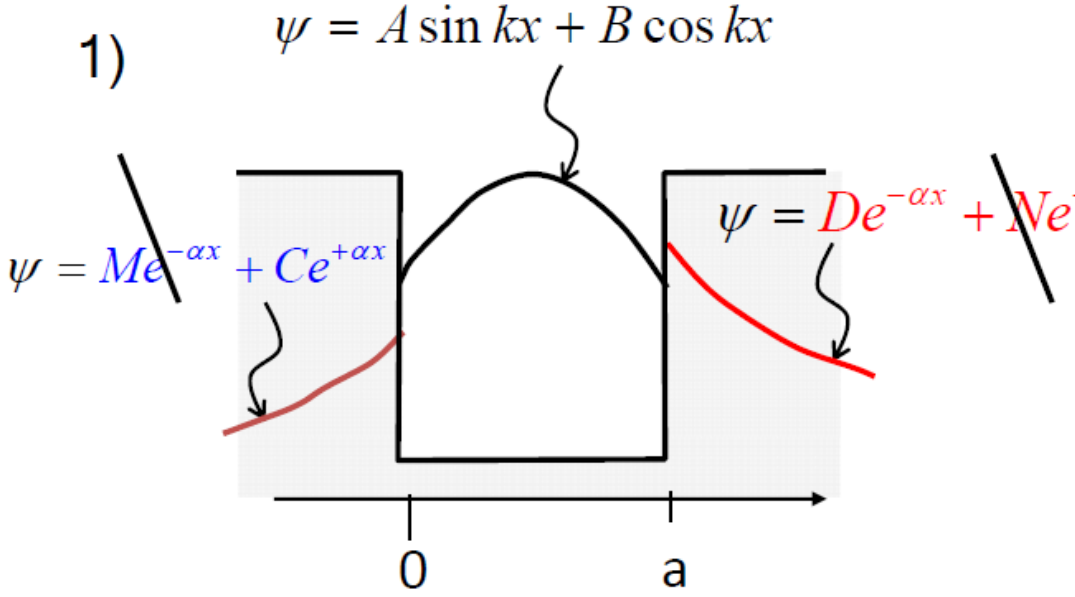


Figure 2.7 | Particle in an infinite potential well: (a) four lowest discrete energy levels, (b) corresponding wave functions, and (c) corresponding probability functions. *(From Pierret [10].)*

Example 3: Finite Potential Well

1)



$\frac{d^2\psi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - V(x))\psi(x) = 0$

$x > a \quad V(x) = V_0 \ (V_0 > E)$

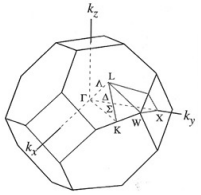
$\psi(x) = D\cdot e^{-\alpha x} + N e^{\alpha x}$

$\alpha = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$

$x < 0$

$\psi(x) = Me^{-\alpha x} + Ce^{\alpha x}$

$M = N = 0$



Example 3: Finite Potential Well

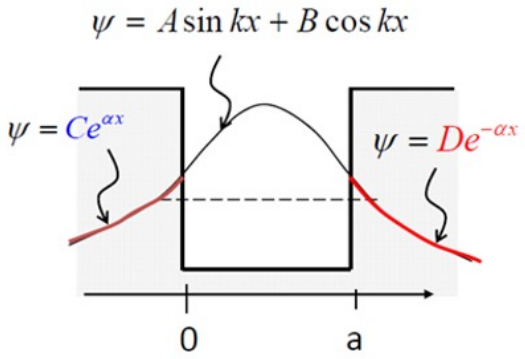
- Boundary Conditions
  - $\varphi(-\infty) = 0$
  - $\varphi(\infty) = 0$
  - $\varphi|_{x=x_B^-} = \varphi|_{x=x_B^+}$
  - $\varphi'|_{x=x_B^-} = \varphi'|_{x=x_B^+}$

$$C = B$$

$$\alpha C = kA$$

$$A\sin(ka) + B\cos(ka) = De^{-\alpha a}$$

$$kA\cos(ka) - kB\sin(ka) = -\alpha De^{-\alpha a}$$



$$\int_{-b}^{+b} -\frac{\hbar^2}{2m}\cdot\psi''(x)\,dx + \int_{-b}^{+b} V(x)\cdot\psi(x)\,dx = \int_{-b}^{+b} E\cdot\psi(x)\,dx$$

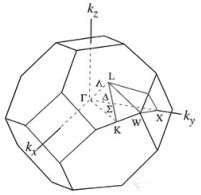
$$\Downarrow$$

$$-\frac{\hbar^2}{2m}\cdot[\psi'(+b) - \psi'(-b)] = 0$$

$$\psi'(+b) = \psi'(-b)$$

- 所以连续

finite $\longrightarrow 0$



Example 3: Finite Potential Well

$$C = B$$

$$\alpha C = kA$$

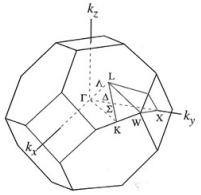
$$A\sin(ka) + B\cos(ka) = De^{-\alpha a}$$

$$kA\cos(ka) - kB\sin(ka) = -\alpha De^{-\alpha a}$$

$$\begin{pmatrix} 0 & 1 & -1 & 0 \\ -k & 0 & \alpha & 0 \\ \sin(ka) & \cos(ka) & 0 & -e^{-\alpha a} \\ \cos(ka) & -\sin(ka) & 0 & \alpha e^{-\alpha a}/k \end{pmatrix} \begin{bmatrix} A \\ B \\ C \\ D \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

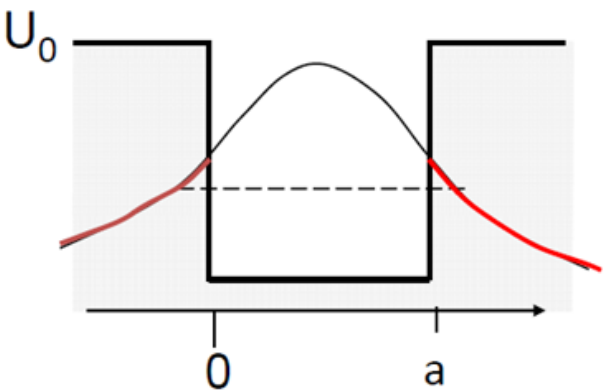
M

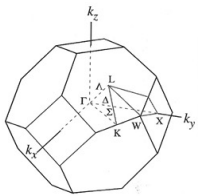
$\det(M) = 0$



Example 3: Bound-level in Finite Well

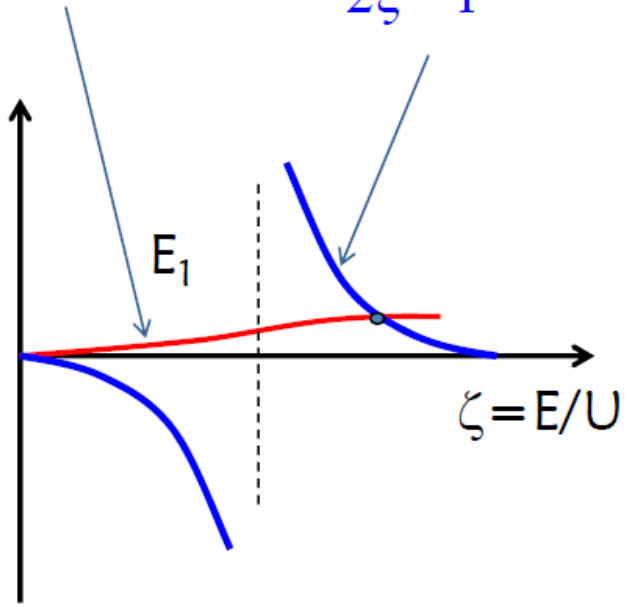
- Det(matrix) = 0
- $\tan(ka) = \frac{2\alpha k}{k^2 - \alpha^2}$ k and α are function of E
- Assume: $\xi \equiv \frac{E}{U_0}$ and $\alpha_0 \equiv \sqrt{\frac{2mU_0}{\hbar^2}}$
- $\tan\left(\alpha_0 a\sqrt{\xi}\right) = \frac{2\sqrt{\xi(1-\xi)}}{2\xi - 1}$
- Only unknown is E
  - Use Matlab function
  - Use graphical method

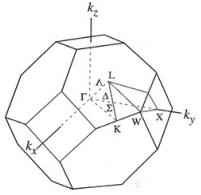




Example 3: Graphical Method for Bound Levels

$$\tan(\alpha_0 a\sqrt{\xi}) = \frac{2\sqrt{\xi(1-\xi)}}{2\xi - 1}$$



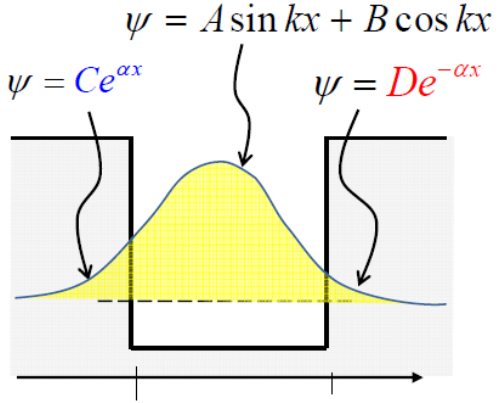


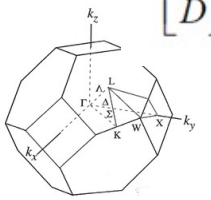
Example 3: Wavefunctions

$$\begin{pmatrix} 0 & 1 & -1 & 0 \\ -k & 0 & \alpha & 0 \\ \sin(ka) & \cos(ka) & 0 & -e^{-\alpha a} \\ \cos(ka) & -\sin(ka) & 0 & \alpha e^{-\alpha a}/k \end{pmatrix} \begin{bmatrix} A \\ B \\ C \\ D \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & \alpha & 0 \\ \cos(ka) & 0 & e^{-\alpha a} \end{pmatrix} \begin{bmatrix} B \\ C \\ D \end{bmatrix} = \begin{bmatrix} 0 \\ -kA \\ -A\sin(ka) \end{bmatrix}$$

$$\begin{bmatrix} B \\ C \\ D \end{bmatrix} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & \alpha & 0 \\ \cos(ka) & 0 & e^{-\alpha a} \end{pmatrix}^{-1} \begin{bmatrix} 0 \\ -kA \\ -A\sin(ka) \end{bmatrix}$$

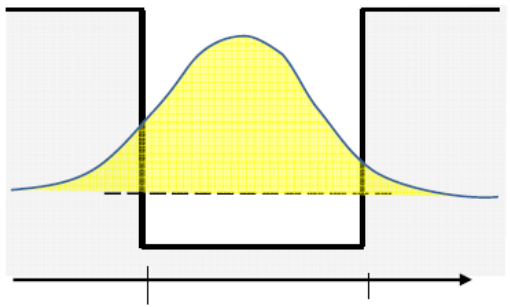




Example 3: Wavefunctions

$$\psi = A\sin kx + B\cos kx$$

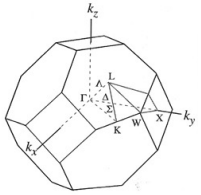
$\psi = Ce^{\alpha x}$ $\psi = De^{-\alpha x}$



$$\begin{bmatrix} B \\ C \\ D \end{bmatrix} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & \alpha & 0 \\ \cos(ka) & 0 & e^{-\alpha a} \end{pmatrix}^{-1} \begin{bmatrix} 0 \\ -kA \\ -A\sin(ka) \end{bmatrix}$$

$$\int_{-\infty}^{\infty} |\psi|^2 dx = 1 \quad \Rightarrow$$

$$\int_{-\infty}^{0} C^2 e^{2\alpha x} dx + \int_{0}^{a} \left[A\sin(kx) + B\sin(kx) \right]^2 dx + \int_{a}^{\infty} D^2 e^{-2\alpha x} dx$$



Five Step for Analytical Solution for General Case

1. $-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\varphi(x) = (E-V)\varphi(x)$ → 2N unknowns for N regions

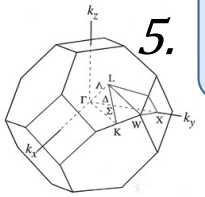
2. $\varphi(\pm\infty)=0$ → Reduces 2 unknowns

3. Boundary condition
   - $\varphi(x)|_{x=x_B^-}=\varphi(x)|_{x=x_B^+}$
   - $\varphi'(x)|_{x=x_B^-}=\varphi'(x)|_{x=x_B^+}$

   → Set 2N-2 equations for 2N-2 unknowns (for finite U)

4. Det(coefficient Matrix) = 0 And find E by graphical or numerical solution

5. $\int_{-\infty}^{\infty}|\varphi(x)|^2dx = 1$ for wave function normalization.



Example 4: Step Potential Function

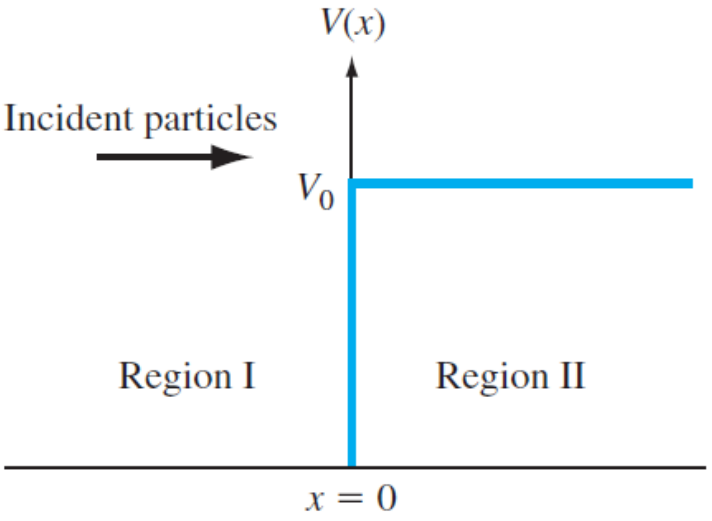


Figure 2.8 | The step potential function.

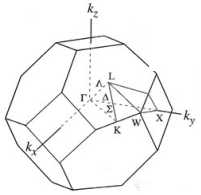
$$\Psi(x) = e^{ikx} + re^{-ikx}, x < 0$$
$$= te^{ik'x}, x > 0$$

$$k = \sqrt{2mE/\hbar^2}$$
$$k' = \sqrt{2m(E - V_0)/\hbar^2}$$

Boundary Conditions:

$\psi(0^-) = \psi(0^+)$

$d\psi/dx|_{0^-} = d\psi/dx|_{0^+}$



Example 4: Step Potential Function

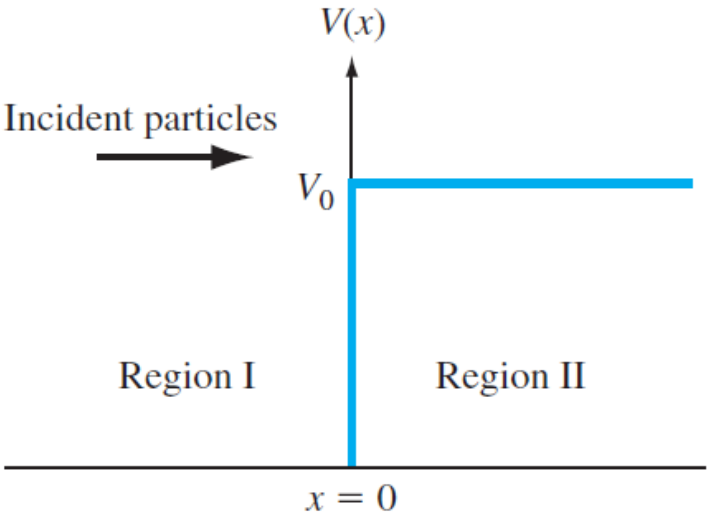


Figure 2.8 | The step potential function.

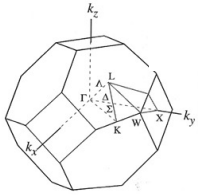
$$\Psi(x) = e^{ikx} + re^{-ikx}, x < 0$$
$$= te^{ik'x}, x > 0$$

$$k = \sqrt{2mE/\hbar 2}$$
$$k' = \sqrt{2m(E - V_0)/\hbar^2}$$

Boundary Conditions:

$$1 + r = t$$
$$k(1 - r) = k't$$

$$t = 2k/(k + k')$$
$$r = (k - k')/(k + k')$$



Example 4: Step Potential Function

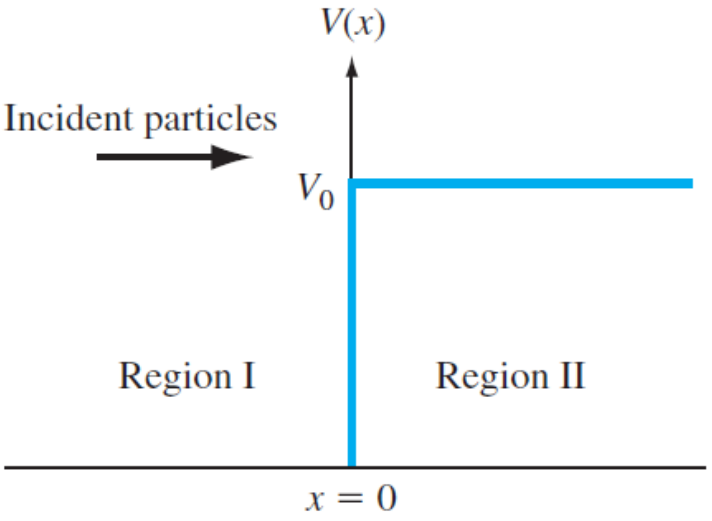
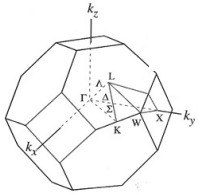


Figure 2.8 | The step potential function.

$$\Psi(x) = e^{ikx} + re^{-ikx}, x < 0$$
$$= te^{ik'x}, x > 0$$

Transmission = current transmitted/current incident
Reflection = current reflected/ current incident

Free particle J = $\hbar k/m|\psi_0|^2$, ψ_0: amplitude
(More correctly, J = Re($\hbar k/m|\psi_0|^2$), in case k is complex), Thus, T = Re(k'$|t|^2$/k), R = $|r|^2$



Repeat for E < V_0

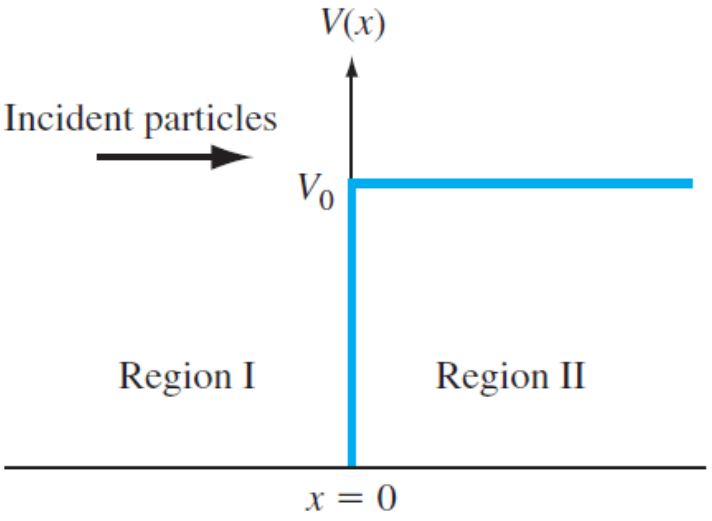


Figure 2.8 | The step potential function.

$$\Psi(x) = e^{ikx} + re^{-ikx}, x < 0$$
$$= te^{ik'x}, x > 0$$

$$k = \sqrt{2mE/\hbar 2}$$
$$k' = \sqrt{2m(E - V_0)/\hbar^2}$$
$$k' = i\eta$$
$$\eta = \sqrt{2m(V_0 - E)/\hbar 2}$$

$J = n \cdot v$
$\downarrow$
$n = |\psi|^2$
$v = Re\{\frac{p}{m}\}$
$\downarrow$
$J = |\psi|^2 \cdot Re\{\frac{\hbar k}{m}\}$

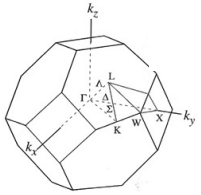
$$t = 2k/(k + i\eta)$$
$$r = (k - i\eta)/(k + i\eta)$$

$R = \frac{J_{ref}}{J_{in}} = |r|^2 = \left|\frac{k - i\eta}{k + i\eta}\right|^2$

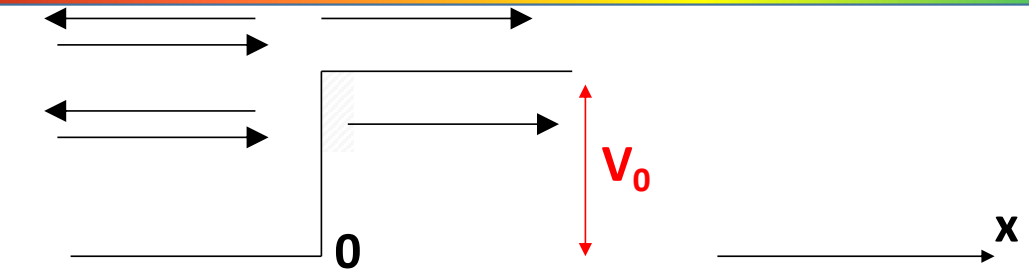
$T = \frac{J_{tran}}{J_{in}} = |t|^2 \cdot Re\{\frac{k'}{k}\}$

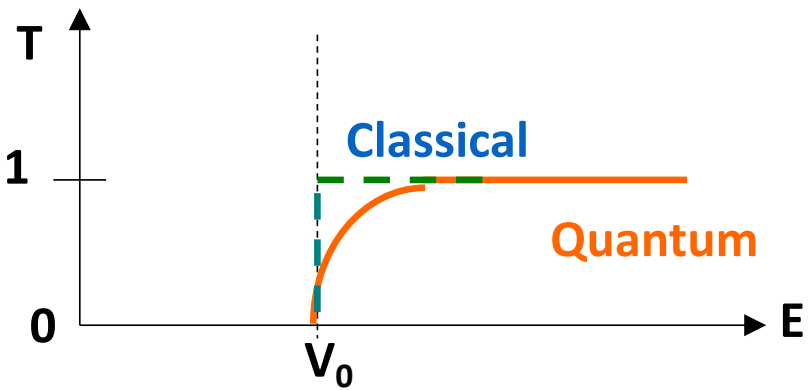
$$T = Re(i\eta|t|2/k) = 0$$
$$R = |r|2 = |k - i\eta|^2/|k + i\eta|^2 = 1$$

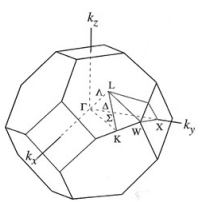
$= \left|\frac{4kk'}{(k+i\eta)^2}\right|$
$\Rightarrow R + T = 1$



Summary: Particle at a step







Example 5: Potential Barrier

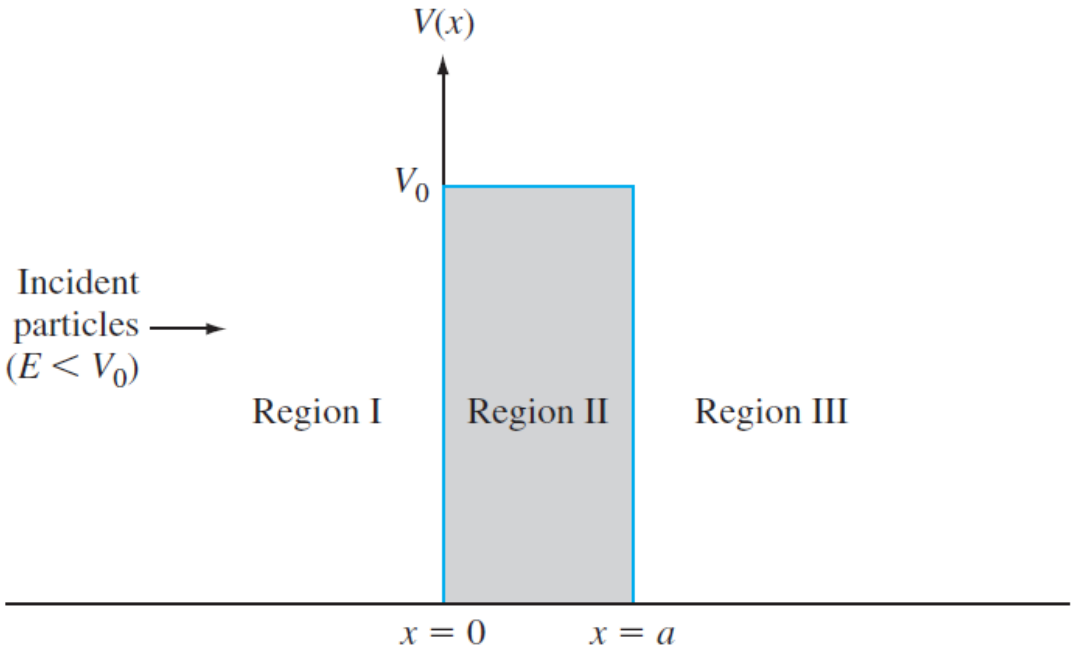
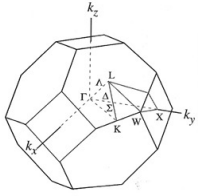


Figure 2.9 | The potential barrier function.



Tunneling

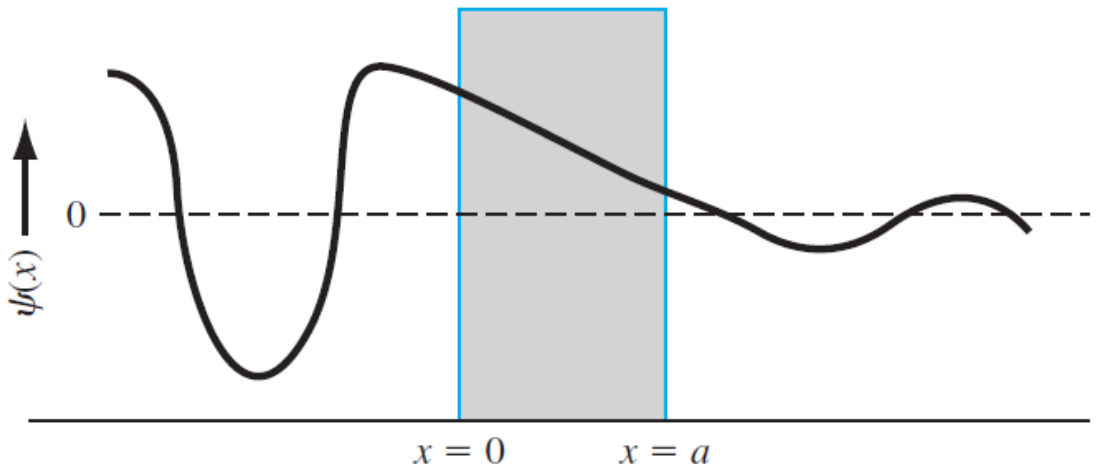
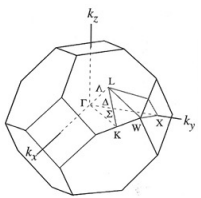
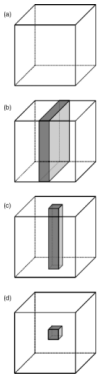


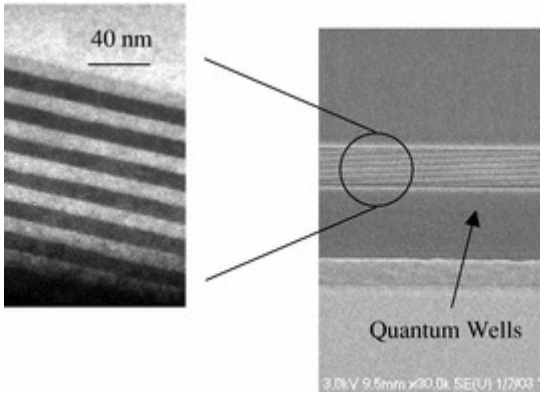
Figure 2.10 | The wave functions through the potential barrier.

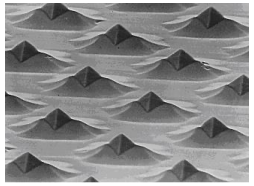


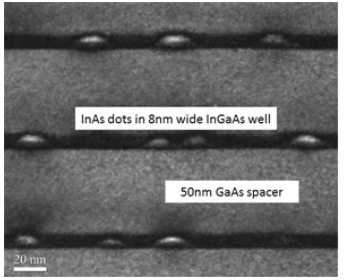
Quantum Confinement

- Confinement in
  - 0-dim: bulk material
  - 1-dim: quantum well
  - 2-dim: quantum wire
  - 3-dim: quantum dot









- Applications:
  - quantum well and quantum dot lasers, quantum cascade lasers, quantum well detectors,
  - High-electron mobility transistors, carbon nanotubes,
  - Nanoparticles.

